# QPath — Supplementary Materials

Companion to **"QPath: An Instrumented Generative-AI Platform for Introductory Quantum Programming Education with Qiskit"**

Anonymous reviewer-oriented supplementary document | Version 1.3 | 13 November 2026

This supplement documents the original pilot cohort, assessment item stems, export schema, descriptive analyses, LLM response ratings, and the verification procedure. It is designed to accompany a six-page IEEE-format manuscript; it is **not** an additional experiment or a validation of teaching effectiveness. No author identifiers or participant-level records are included. **No source code or replay scripts are attached to or hosted with this supplement.**

**Evidence note.** Figures, methods, and interpretation refer to the companion manuscript. Quantities marked "recomputed" were recomputed by the authors from the 7 November 2026 research export; operational prompt wording, complete test distractors, and the original consent wording were not available in that export.

## S1. Pilot cohort, protocol, and reconstruction

The original study used one arm with no contemporaneous control. Learners registered, affirmed electronic study consent, completed a pre-test, accessed the guided Qiskit learning path and optional AI Tutor support, then completed a post-test and survey. A complete case required consent, a pre-test, at least one lesson activity, at least one AI interaction, a post-test and survey completion. The flow below reports records, not randomized allocation.

| Milestone | Count | Interpretation |
|---|---|---|
| Accounts registered by the cutoff | 24 | Researcher-reconstructed registration cohort |
| Pre-tests submitted | 21 | 87.5% of registered accounts |
| Post-tests submitted | 8 | All eight also completed the survey |
| Pre-test without post-test | 13 | Other pre-test completers; not necessarily zero lesson exposure |
| Complete pre/post/survey cases | 8 | 38.1% of the 21 pre-test completers |
| Pre-to-post noncompletion | 61.9% | 13/21; possible selection bias |

Cohort reconstruction used a retrospective time boundary: registration and AI interaction timestamps strictly before 1 July 2026 (i.e., through 30 June). The later 7 November export also contains two additional accounts and five later AI interactions. These later records are excluded from the published original-cohort analyses. The boundary was applied during reconstruction; advance specification is not documented.

The eight complete cases had a baseline pre-test mean of 53.75% (SD 31.14), compared with 45.38% (SD 21.84) among 13 other pre-test completers. This 8.37 percentage-point descriptive difference was not significance-tested. Since complete-case status includes having used AI, the paired outcome estimate may overrepresent engaged learners.

## S2. Assessment instruments and survey mapping

The source export contains exactly 10 unique pre-test question stems and 10 unique post-test stems for the original cohort, each with a keyed correct response and a short explanation. It does **not** contain the complete list of multiple-choice distractors as presented to students, and formal form-equivalence validation was not documented. Below are the original exported stems and correct responses, not a newly generated instrument.

### Pre-test: ten original item stems

**`pre_q1` — Quantum circuit**
In Qiskit, what does a QuantumCircuit(1, 1) usually represent?
**Key:** A circuit with one quantum bit and one classical bit

**`pre_q2` — Qubit, state, and measurement**
Before any gate is applied, a newly created qubit in a basic circuit is usually initialized to:
**Key:** |0>

**`pre_q3` — Hadamard gate**
What is the main role of a Hadamard gate H applied to |0>?
**Key:** It creates a superposition state

**pre_q4 — Measurement**
What does measurement do in an introductory quantum circuit?
**Key:** It maps the quantum state to a classical outcome

**pre_q5 — Shots and counts**
Why are circuits often executed with many shots?
**Key:** To estimate the distribution of possible measurement outcomes

**pre_q6 — CNOT gate**
In a CNOT gate, the target qubit is flipped when:
**Key:** The control qubit is in state 1

**pre_q7 — Classical vs quantum**
Which statement best distinguishes a qubit from a classical bit before measurement?
**Key:** A qubit can be described by amplitudes and may be in superposition

**pre_q8 — Qiskit syntax**
In Qiskit, which line applies a Hadamard gate to qubit 0?
**Key:** qc.h(0)

**pre_q9 — Entanglement intuition**
A common purpose of using H on one qubit followed by CNOT is to demonstrate:
**Key:** Correlated quantum outcomes

**pre_q10 — Debugging**
A frequent beginner mistake in Qiskit measurement is:
**Key:** Measuring into a classical bit that was not allocated

### Post-test: ten original item stems

**post_q1 — Quantum circuit**
What are the two resources specified in QuantumCircuit(2, 2)?
**Key:** Two qubits and two classical bits

**post_q2 — Qubit, state, and measurement**
If a qubit remains in |0> and is measured many times, what result is expected?
**Key:** Mostly 0

**post_q3 — Hadamard gate**
After applying H to |0> and measuring many shots, the counts should be approximately:
**Key:** A balance between 0 and 1

**post_q4 — Measurement**
Why is a classical bit used in qc.measure(0, 0)?
**Key:** To store the measurement outcome of qubit 0

**post_q5 — Shots and counts**
If a result dictionary is {'0': 510, '1': 490}, what does it suggest?
**Key:** An approximate 50/50 outcome distribution

**post_q6 — CNOT gate**
In a simple Bell-style circuit, why can outcomes 00 and 11 appear more often than 01 and 10?
**Key:** Because the qubits are correlated by H and CNOT

**post_q7 — Classical vs quantum**
Why should a learner avoid saying that a superposed qubit is simply 'both classical values at once'?
**Key:** Because amplitudes and measurement probabilities are more precise than that simplified phrase

**post_q8 — Qiskit syntax**
Which instruction measures qubit 0 into classical bit 0?
**Key:** qc.measure(0, 0)

**post_q9 — Entanglement intuition**
Which pair of operations is commonly used to introduce correlated two-qubit outcomes?
**Key:** H then CNOT

**post_q10 — Debugging**
A student writes QuantumCircuit(1, 0) then qc.measure(0, 0). What is the likely issue?
**Key:** No classical bit was allocated for the measurement result

**Assessment scoring:** each original pilot form comprised ten items and yielded scores in ten-point percentage increments. Several item pairs probe the same concept (for example `pre_q8`/`post_q8` and `pre_q10`/`post_q10`); this supports the parallel-form design but means some pre-to-post change may reflect re-exposure to similar items. Full answer-option sets and test-form administration screens must be retrieved separately from an archived application version for a complete operational instrument.

### Learner survey: available variable labels

The export provides seven named 1–5 numerical fields but does not preserve the exact wording shown to participants. Labels below are derived only from field names. Eight survey records belong to the eight complete cases.

| Survey variable label | Mean (1–5) |
|---|---|
| Scaffolding clarity | 3.875 |
| Qiskit examples | 3.875 |
| AI feedback | 3.875 |
| Exercise generation | 3.125 |
| Reflection prompts | 3.625 |
| Over-reliance awareness | 3.375 |
| Usability | 3.875 |

Overall seven-item survey mean = 3.6607/5. Interpret as self-report only.

## S3. LLM instruction and prompt evidence

The prototype AI Tutor used Llama-3.1-8B-Instant through the Groq API, with support modes including conceptual explanation, guided hints, practice generation, feedback, reflection and a local curated-template fallback. Context variables described in the manuscript include current module, target concept, learner attempt text and requested support type. Concept Builder events are action logs rather than additional LLM model responses. In the original cohort, 14 events consisted of seven `concept_builder_request` entries and seven recorded follow-up actions (three visual cards, one mini-quiz, one misconception check, one analogy, and one simpler explanation). The two `curated_template` output records in the 90-row AI-interaction stream came from the Concept Builder: one mini-quiz and one misconception check (`model=concept-builder-v12.4`). The action log links these two curated responses to their corresponding events; therefore the streams overlap and must not be summed.

### Interpreting the support modes and activity features

**Explain Simply** requests a less technical restatement of a selected concept or explanation, while **Ask Me a Question** requests a guiding or comprehension-check question. Neither mode was evaluated as a separate experimental condition. The exercise-generation feature offers a lesson-context practice prompt rather than a formally validated item bank; seven LLM calls in the original cohort carry the exact task label `Generate a practice exercise`. These are request counts, not automatically validated exercises or successful learning outcomes.

The **Local Fallback** is a rule/template path intended for cases in which external generation is unavailable or fails. Its activation flag was zero across all 90 original-period AI interactions. The two Concept Builder outputs classified as `curated_template` must not be misreported as fallback activations. Function availability in the platform architecture is distinct from observed use and pedagogical effectiveness.

### Illustrative prompt structure from the main manuscript

> You are a quantum programming tutor. The learner is working on the current module: [module name]. The target concept is: [concept]. The learner attempt is: [learner text]. Provide a short hint or guiding question without giving the final answer directly. If Qiskit code is relevant, connect the explanation to a minimal Qiskit example.

**Evidence note.** This quoted prompt is illustrative, not a verified byte-for-byte historical prompt. The available research export has prompt-template metadata but intentionally excludes live raw prompts and replies. Exact deployed system-prompt files and prompt versions should only be archived if retrievable and verified from source code.

## S4. LPQS scoring, distribution, and uncertainty

One evaluator rated 42 responses from nine learners (five also belonged to the eight complete cases). All rated model replies came from Llama-3.1-8B-Instant. Each response received integer scores from 1 to 5 on seven criteria. LPQS was calculated as the arithmetic mean of the seven ratings for a response, then averaged descriptively over rated responses. This approach does not establish interval-scale measurement validity or inter-rater reliability.

| Criterion | Mean | Median | SD (sample) |
|---|---|---|---|
| Conceptual accuracy | 3.60 | 3 | 0.86 |
| Answer relevance | 3.45 | 3 | 0.74 |
| Pedagogical clarity | 3.24 | 3 | 0.48 |
| Scaffolding quality | 3.21 | 3 | 0.52 |
| Qiskit alignment | 3.71 | 3 | 0.97 |
| Reflection support | 3.17 | 3 | 0.44 |
| Personalization | 3.19 | 3 | 0.51 |

| Rating profile group | N | Mean LPQS | Interpretation |
|---|---|---|---|
| All seven criteria rated 3 | 27 | 3.00 | 64.3% of rated responses |
| At least one criterion not 3 | 15 | 4.03 | Descriptive subgroup; not a second rater |
| All rated responses | 42 | 3.37 | Overall median 3.00 |

The platform interface described in the manuscript has rating sliders initially set to 3. The export cannot distinguish deliberate midpoint judgments from unchanged defaults, nor prove that a full transcript was opened for each historical rating. None of the 42 recorded criterion scores is below 3. The selection of the 42 responses for rating was not probability-based or documented as random. Scores may be correlated within learners. Accordingly, LPQS is a preliminary descriptive rubric application, not a validated measure of tutor quality.

## S5. Statistical analyses and how they were checked

All calculations below use the eight matched complete cases from the original cohort. Primary analysis is descriptive, and inferential statistics are exploratory due to $n = 8$, selection through completion, the absence of a control condition, and the short unvalidated tests.

| Quantity | Computation | Recomputed result |
|---|---|---|
| Mean raw paired change | mean(post - pre) | 30.00 percentage points |
| Mean pre-test | 8 matched observations | 53.75% |
| Mean post-test | 8 matched observations | 83.75% |
| Class-average normalized gain | (83.75 - 53.75) / (100 - 53.75) | 0.64865 (reported 0.65) |
| Paired t-test | df=7, two-sided | t=2.5752, p=0.03673 |
| 95% t CI for mean change | 30 +/- t(0.975,7) x (SDdiff/sqrt(8)) | [2.45, 57.55] pp |
| Cohen dz | mean difference / SD of differences | 0.9105 |
| Approx. Hedges gz | dz x [1 - 3/(4*7-1)] | 0.8093 |
| Cohen dav | 30 / mean(SDpre, SDpost) | 1.3586 |
| Wilcoxon signed-rank | W=2; 7 nonzero pairs; two-sided exact | p=0.046875 (paper: 0.047) |
| Pre/post Pearson correlation | Pearson r across 8 matched cases | 0.066 |

Both reported $p$-values are just below 0.05; they do not provide independent confirmation or causal evidence. Two pre-test scores were 100%, one paired change was zero, one was negative, and six were positive. Because a learner at 100% cannot improve, the two ceiling learners account for the zero and the negative change, and all six positive changes come from learners below the ceiling. The group-level Hake gain of means is defined despite individual ceiling cases. Reporting the average of individual normalized gains would be inappropriate here because individual gain is undefined when pre-test is 100%.

**How these values were checked:** the authors recomputed the values above from the researcher-held research export. The aggregate figures (means, SDs, and the test statistics) can be re-derived by a reviewer from the table above. The Wilcoxon statistic requires the eight individual paired differences, which are held privately and are not released. **No script or code package accompanies this supplement.**

## S6. Instrumentation and data dictionary

The platform research export is a multi-table workbook with CSV copies. In the 7 November snapshot, the later, evolving platform export contains 26 accounts and 95 AI interactions; the original pilot uses only records within the retrospective window. Particular tables supply distinct evidence types:

| Table | Relevant fields / role | Caveat |
|---|---|---|
| progress_summary | Registration state, pre/post flags, case completeness | AI interaction count is a later snapshot counter |
| test_attempts | Attempt type, score, total, keyed response map | No participant code in this table |
| question_responses | Question stems, keyed correct responses, item scores | Rows link to participants and timestamps; withheld |
| ai_interactions | Timestamp, mode, provider/model, task, latency | 90 original period rows after filtering |
| llm_evaluations | Seven LPQS criteria; link to interaction time | Some ratings entered after original period |
| survey_item_scores | Seven 1–5 numerical items | Exact original question wording not in export |
| consent_records | Consent-version field and recorded timestamp | Does not contain full text of informed consent |
| ai_request_timing_parsed | Delay before AI request | Only 12 original-period observed timings |
| learner_attempts | Structured attempts / text if available | Zero rows in provided export |
| concept_builder_events | Builder action logs | 14 original-period events, not 14 AI replies |

### Event-count reconciliation for the original cohort

| Export stream / status | Count | Interpretation |
|---|---|---|
| AI interactions, total | 90 | 88 model calls plus two curated-template response records |
| Concept Builder events | 14 | Seven request events and seven action events; separate logging stream |
| Explicit practice-exercise requests | 7 | LLM calls with task label `Generate a practice exercise`; output correctness untested |
| Simulator opened | 24 | Event occurrences, not distinct learners or verified execution |
| Animation viewed | 24 | Event occurrences, not verified learning outcomes |
| Simulator completed | 2 | Learner-reported activity completion, not an audit of simulator output |
| Local fallback used | 0 | No fallback activation among the 90 original-cohort AI interactions |
| Timed AI requests | 12 | Only these events support mean time-before-request of 169.3 s |

The original-period AI interactions comprise 88 LLM calls and two curated-template outputs (Concept Builder mini-quiz and misconception check), both linked to matching action records. The 14 Concept Builder event logs represent a separate, partly overlapping action stream and must not be added to the 90 interactions. Account-level usage median was 2, with five accounts having zero interaction and a top account contributing 20/90 (22.2%). The timing mean of 169.275 s derives from 12 observations, not the full interaction log.

## S7. Ethics, privacy, availability, and release restrictions

The pilot was conducted without referral to an institutional research ethics committee; no formal committee approval or exemption determination was obtained. The platform required electronic study consent at initial registration, and the export includes dated consent-version records (v1). The exact consent-screen wording is not included in the available export. An institutional review of the ethical status, informed-consent scope, and permitted data-sharing terms remains appropriate before a public data release. No unverified exemption is claimed.

The strict research export excludes direct names and high-risk raw free text, but retained stable pseudonymous codes, event times, and linkable observations in the researcher-held files. These are not equivalent to irreversibly anonymous public data. The anonymous repository currently supplies this supplementary PDF with aggregate statistics, question stems and keys (but no personal answers), and instrument variable labels. The platform source, executable replay code, and participant-level export are not hosted there. The item stems/keys are reviewer-only material and may require separate permission before public posting or reuse in teaching.

**Excluded from public or anonymous review artifacts:** raw participant rows, per-person test scores and ratings, stable participant codes, operational transcripts, specific timestamps, participant-code crosswalk, database URLs, provider keys, raw consent records, file names or paths from the researchers' systems, and identifiable feedback. Authorship identifiers must also be withheld if anonymous review applies.

The ISPA 2026 webpage specifies a six-page IEEE-format PDF for the main manuscript but does not unambiguously authorize supplementary uploads. Confirm that CMT supports additional files before submitting this supplement. Public DOI publication should follow, not precede, an anonymity and privacy review.

**Expected outcomes of the authors' local check:** $n = 24$ accounts, 21 pre-tests, 8 post-tests, 90 original AI interactions, 42 rated LLM responses, normalized gain 0.64865, paired $t$ $p = 0.03673$, Wilcoxon $p = 0.046875$, and LPQS mean 3.36738. These values were reproduced by the authors in a local check against the private export; the checking procedure is not distributed.

**Source and status:** Calculations derive from the QPath manuscript and the 7 November 2026 research export/codebook; the gain-of-means definition follows Hake (1998). Event counts in S3/S6 were verified by the authors against the cutoff-filtered export; interface-mode descriptions follow the manuscript and were not independently established by the exported rows. No new experiments or raters were introduced.